1 Lecture 01 - Introduction to CS 374

Important Terms

1.1 Alphabets

Let Σ be an **alphabet**, an arbitrary **finite** set of **symbols** or **characters**

Examples include:

- $\Sigma = \{A, B, \ldots, Z\}$
- $\Sigma = \{0, 1\}$

1.2 Strings

A **string** or **word** is a **finite** sequence of symbols from an alphabet

Formally speaking, w is a string over Σ if it is one of

- the **empty string**
- an ordered pair (a, x), where $a \in \Sigma$ and x is a string over Σ

1.2.1 Example

$$\begin{aligned} STRING &= (S, (TRING)) \\ &= (S, (T, (RING))) \\ &= \ldots \\ &= (S, (T, (R, (I, (N, (G, \varepsilon)))))) \end{aligned}$$

1.3 Functions

Functions are defined recursively, where length $|w|$ represents the number of symbols in w

$$|w| := \begin{cases} 0 & \text{if } w = \varepsilon \\ 1 + |x| & \text{if } w = a \cdot x \end{cases}$$

1.3.1 Example

$$\begin{aligned} |HAT| &= 1 + |AT| \\ &= 1 + (1 + |T|) \\ &= 1 + (1 + (1 + |\varepsilon|)) \\ &= \boxed{3} \end{aligned}$$

1.4 Concatenation

Let **concatenation**, $w \cdot z$, of strings w and z be represented as

$$w \cdot z := \begin{cases} z & \text{if } w = \varepsilon \\ (a, x \cdot z) & \text{if } w = a \cdot z \end{cases}$$

1.4.1 Example

$$\begin{aligned} NOW \cdot HERE &= N \cdot (OW \cdot HERE) \\ &= N \cdot (O \cdot (W \cdot HERE)) \\ &= N \cdot (O \cdot (W \cdot (\varepsilon \cdot HERE))) \\ &= N \cdot (O \cdot (W \cdot HERE)) \\ &= NOWHERE \end{aligned}$$

1.5 Induction

- Consider an arbitrary object (probably of size n)
- **Explicitly** state a **strong** inductive hypothesis (i.e. `for all k < n`
- Explicitly handle each case for the object
- Use inductive hypothesis when its useful
- End by pointing out you handled all cases

1.5.1 Example